

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

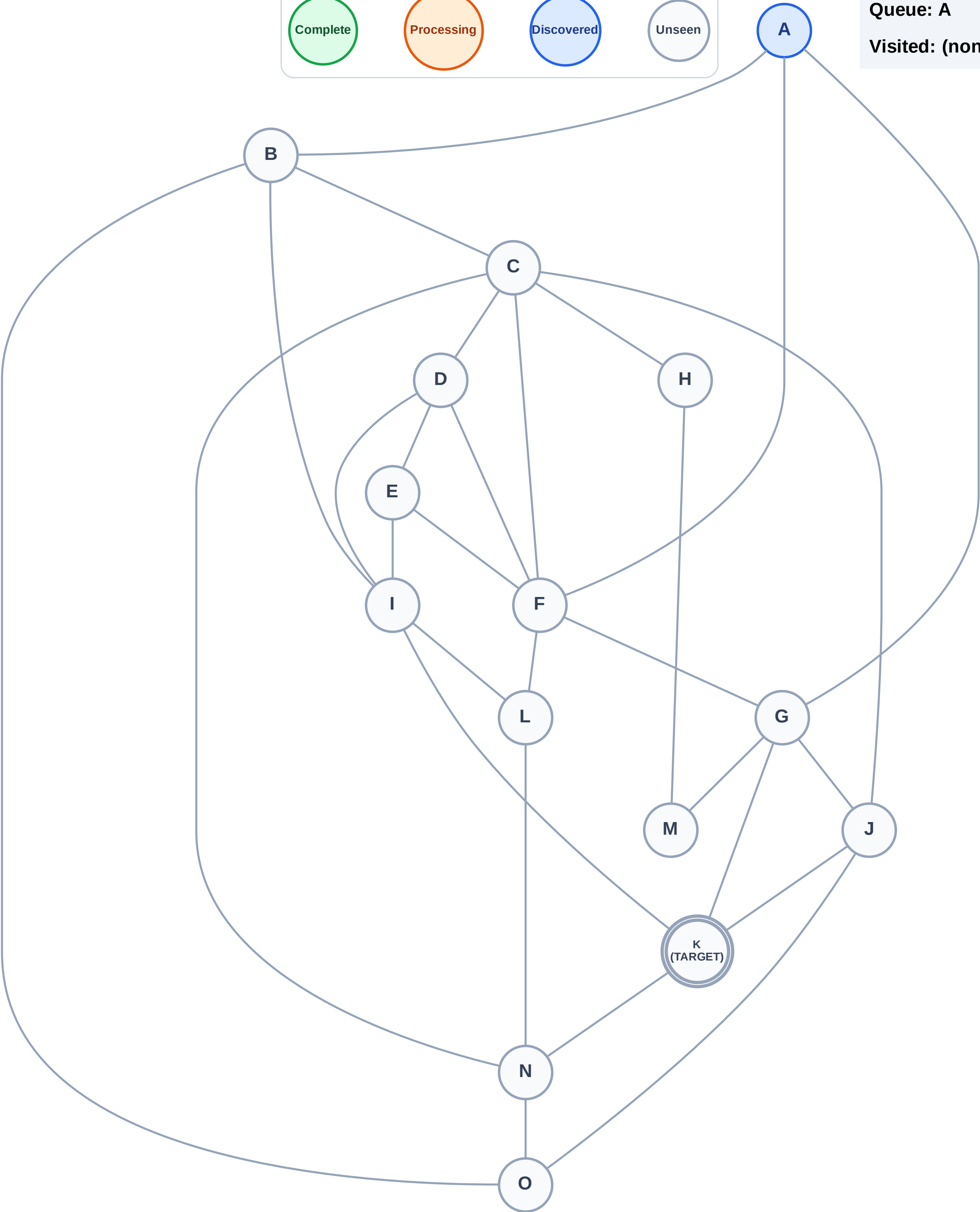
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

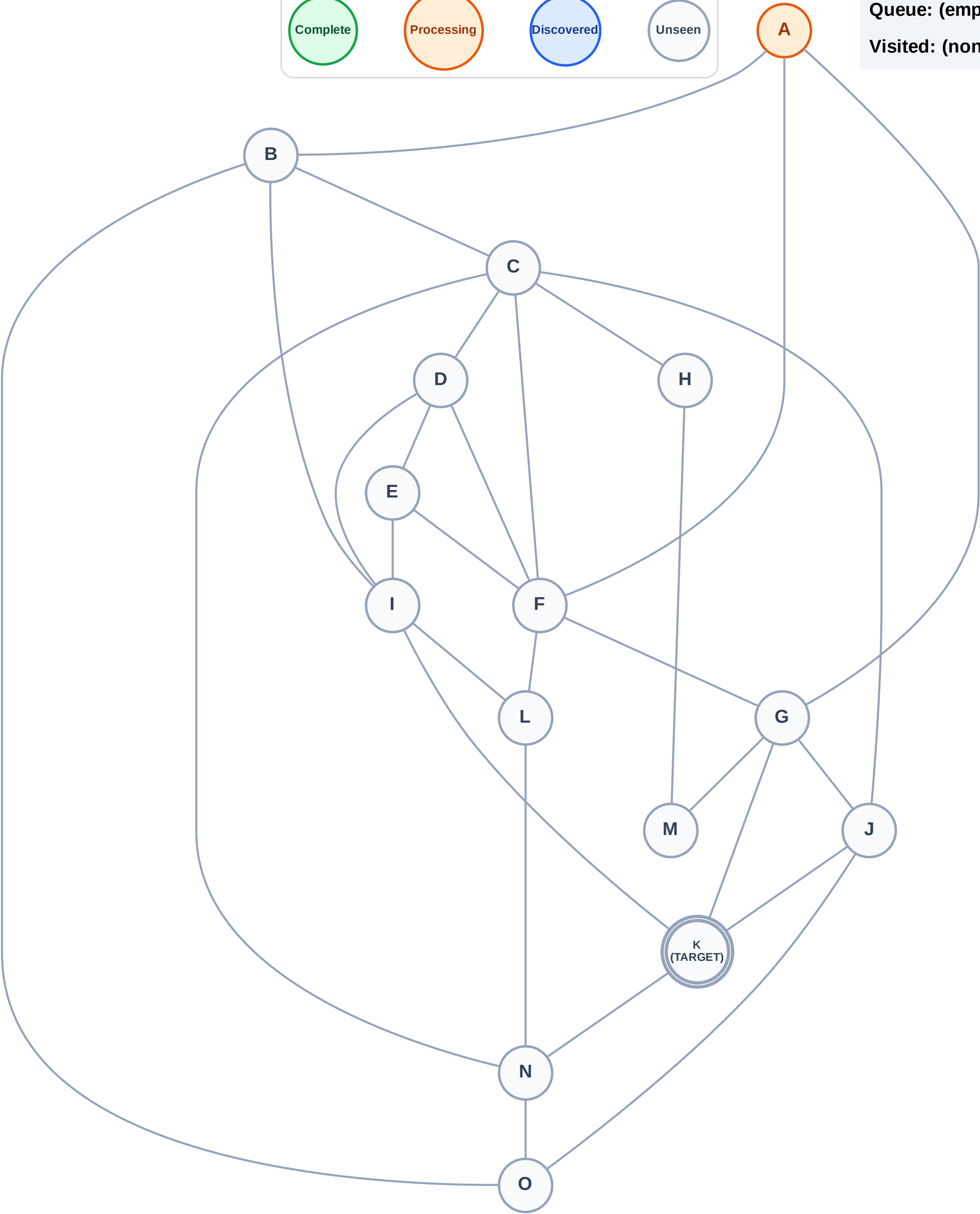
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

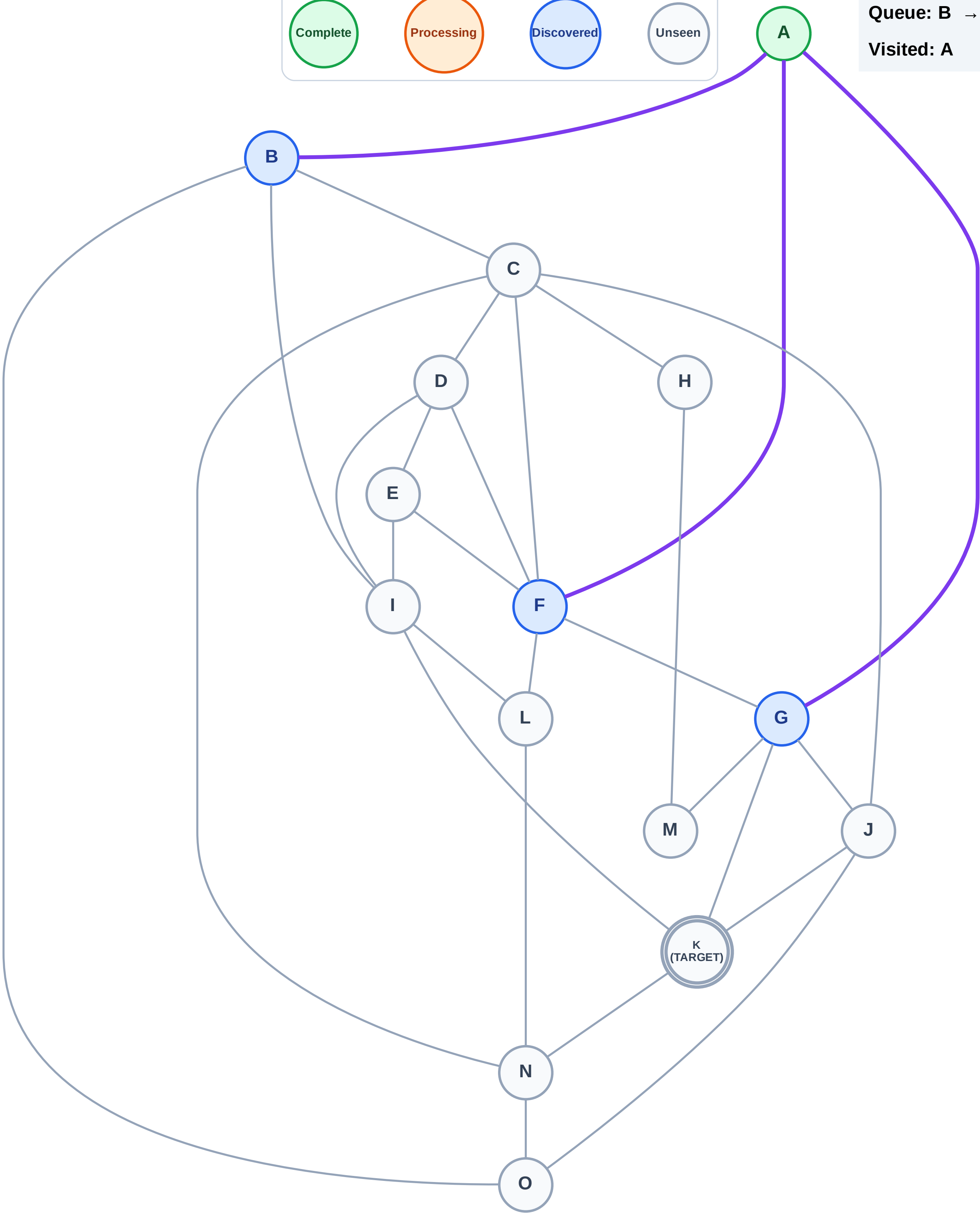
Step 3: Discover B, F, G from A

Legend



Queue: B → F → G

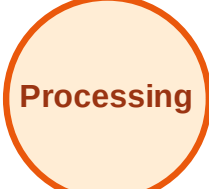
Visited: A



BFS Traversal

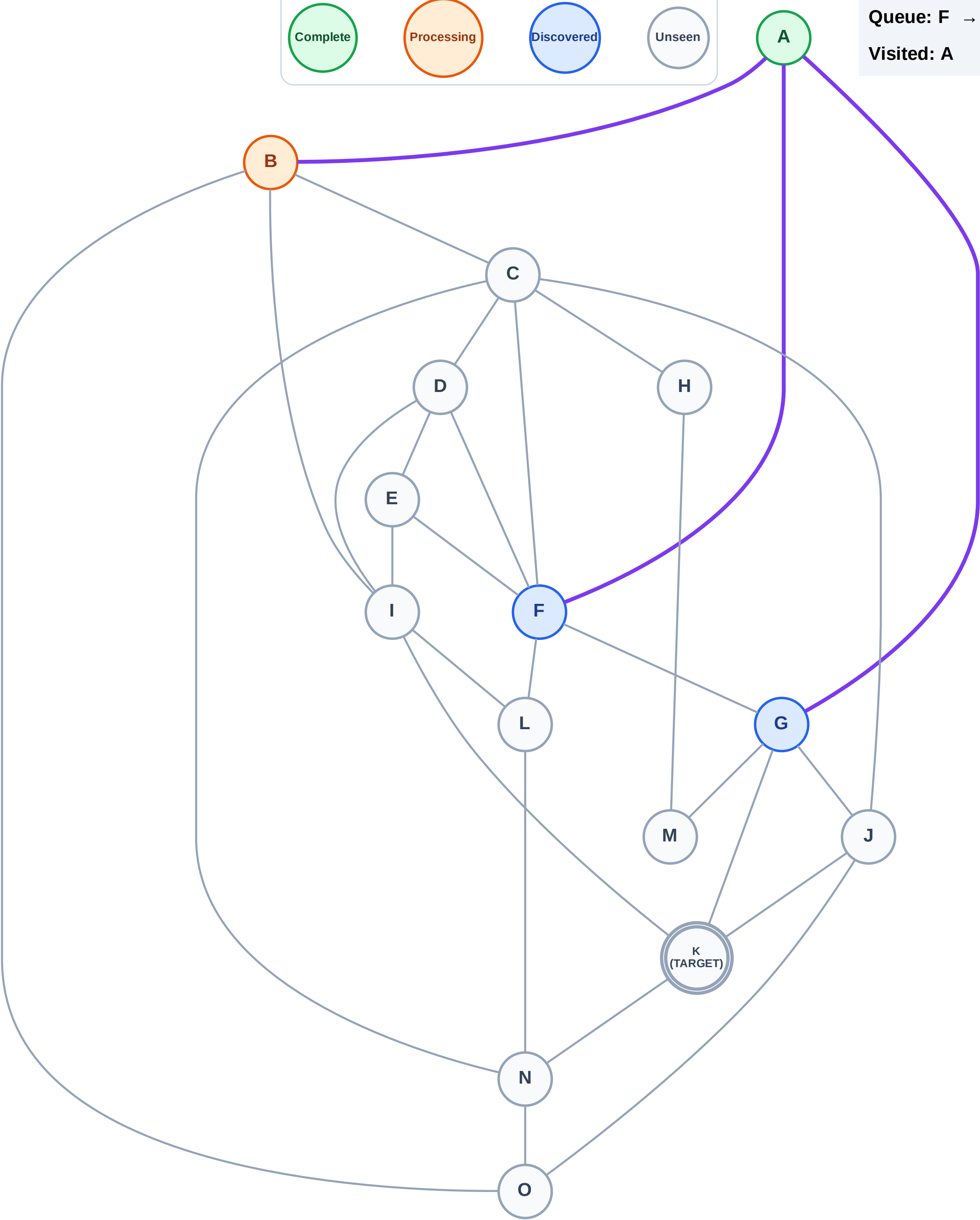
Step 4: Process node B

Legend



Queue: F → G

Visited: A



Step 5: Discover C, I, O from B

Step 5: Discover C, I, O from B

Legend

Complete

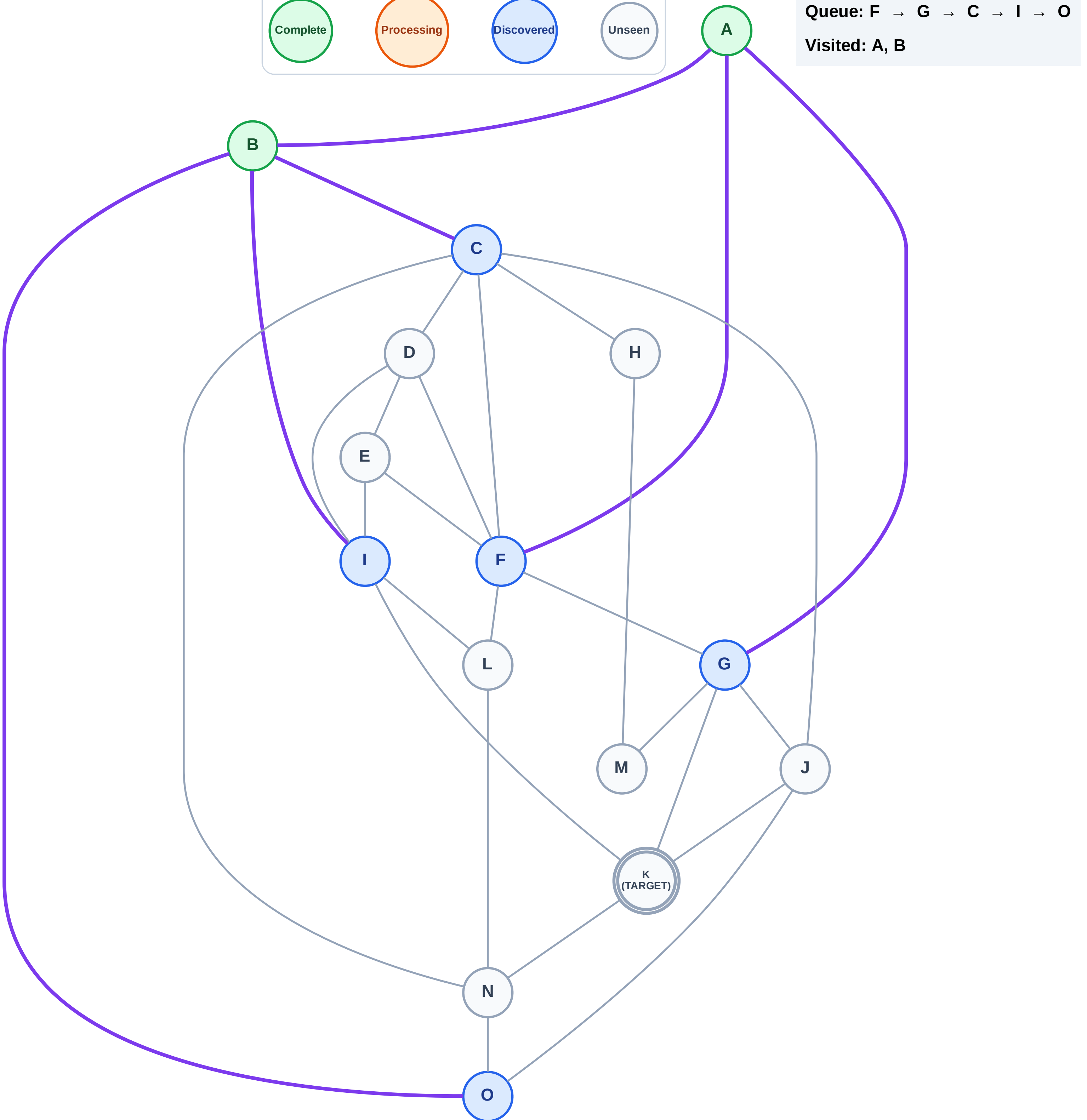
Processing

Discovered

Unseen

Queue: F → G → C → I → O

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

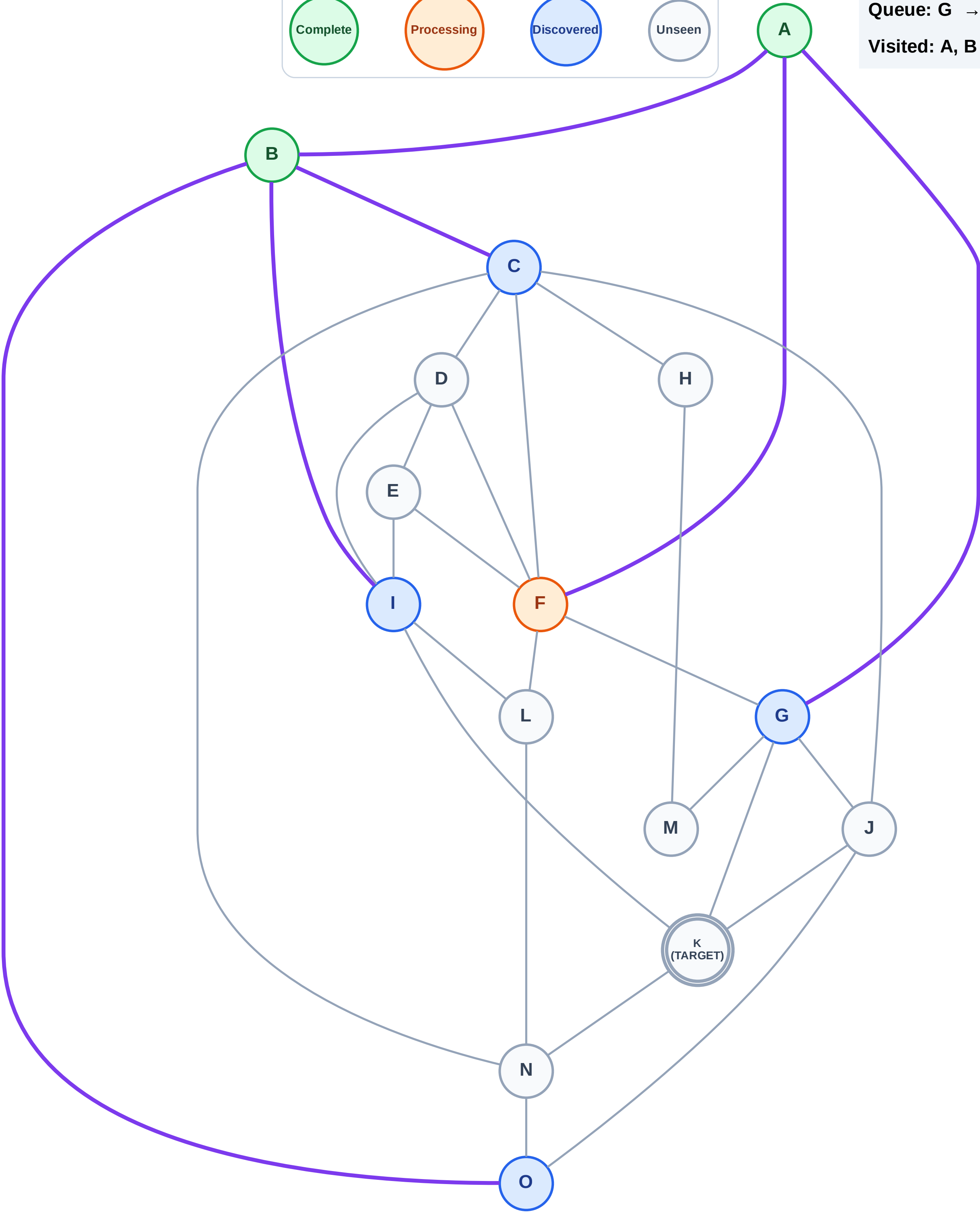
Processing

Discovered

Unseen

Queue: G → C → I → O

Visited: A, B



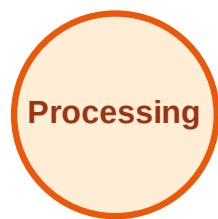
BFS Traversal

Step 7: Discover D, E, L from F

Legend



Complete



Processing

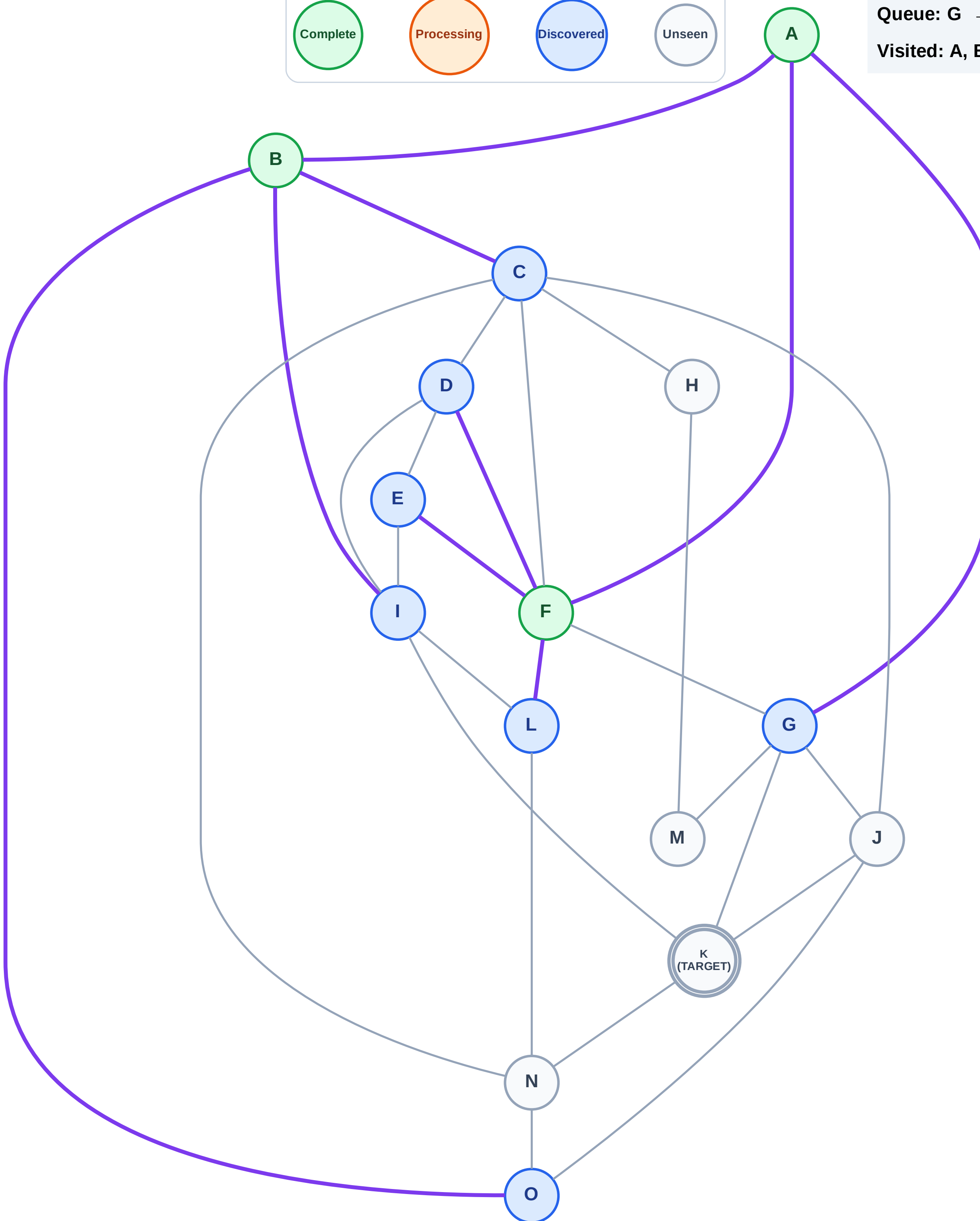


Discovered



Unseen

Queue: G → C → I → O → D → E → L
Visited: A, B, F



Step 8: Process node G

Step 8: Process node G

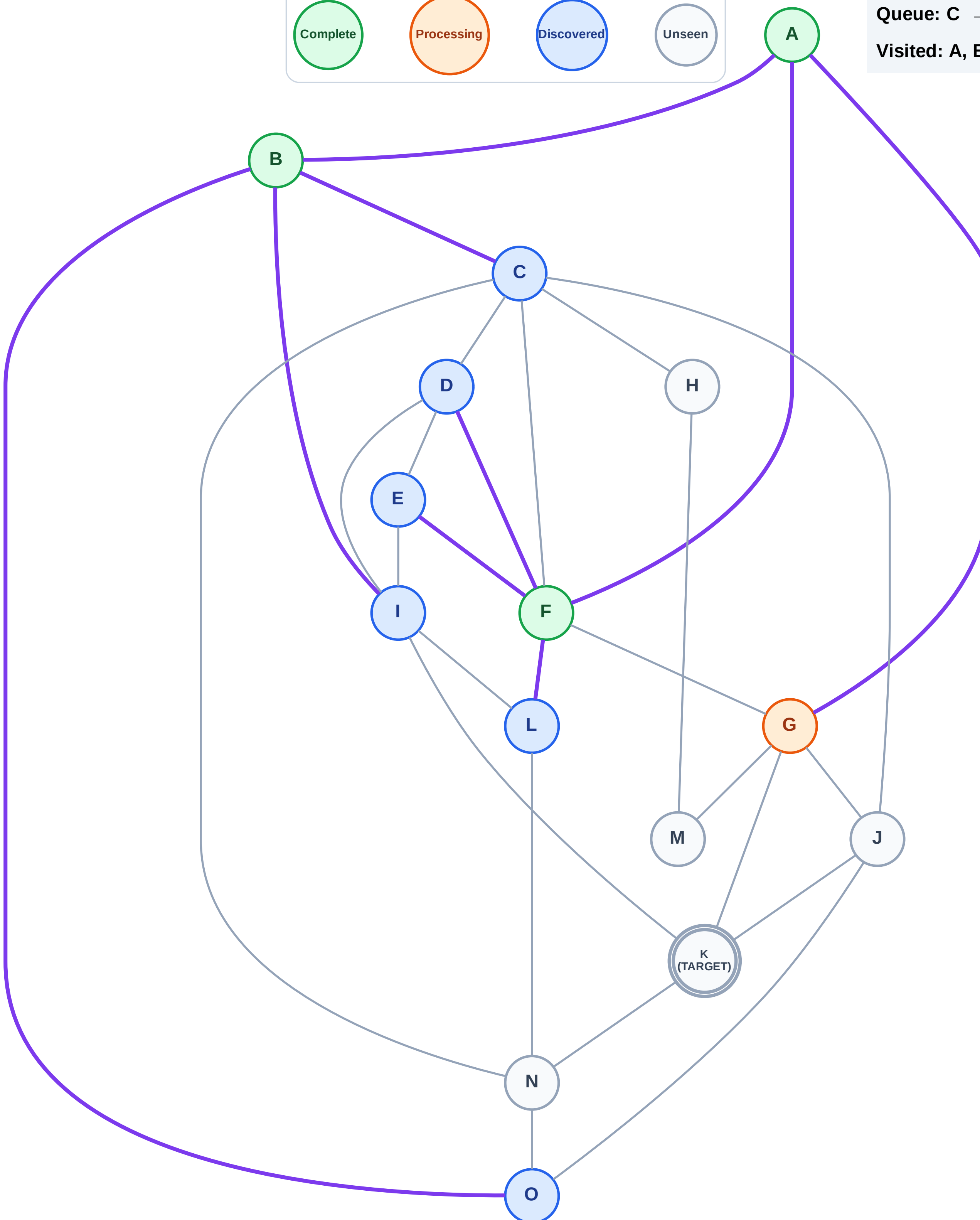
A diagram showing four stages of data processing, each represented by a colored circle with a corresponding label inside:

- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)

Processing

Unseen

Visited: A, B, F



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend

Complete

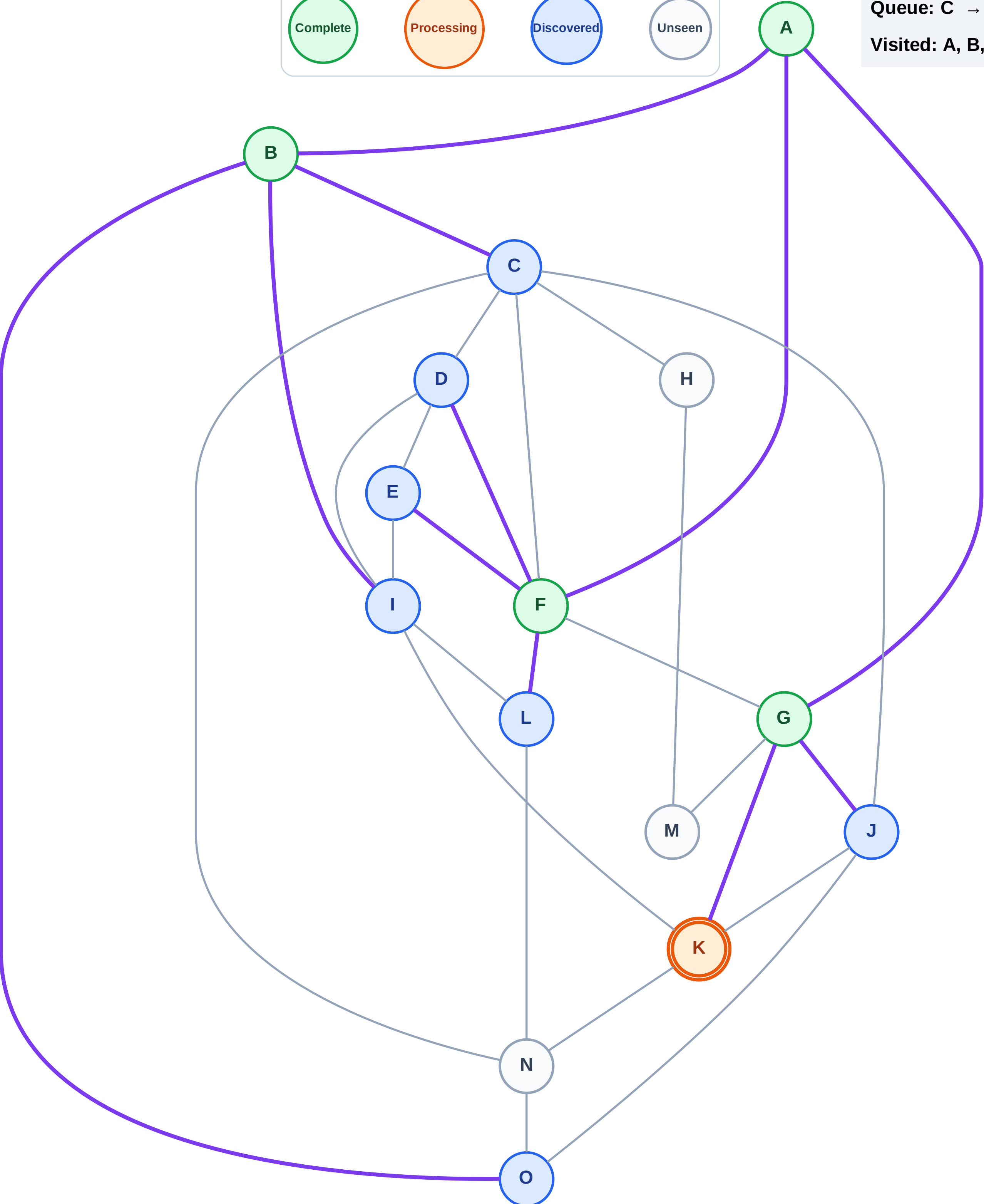
Processing

Discovered

Unseen

Queue: C → I → O → D → E → L → J

Visited: A, B, F, G, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

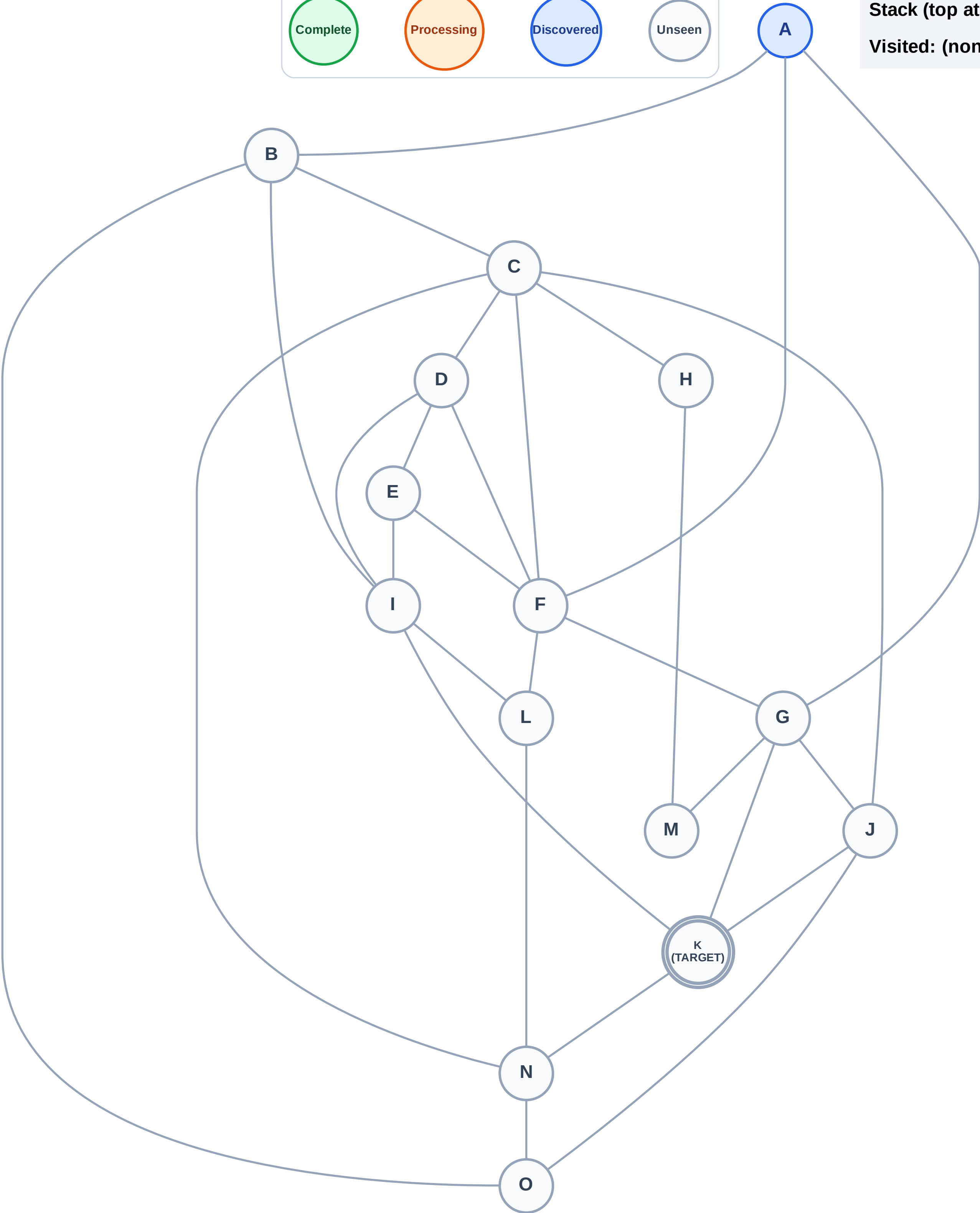
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

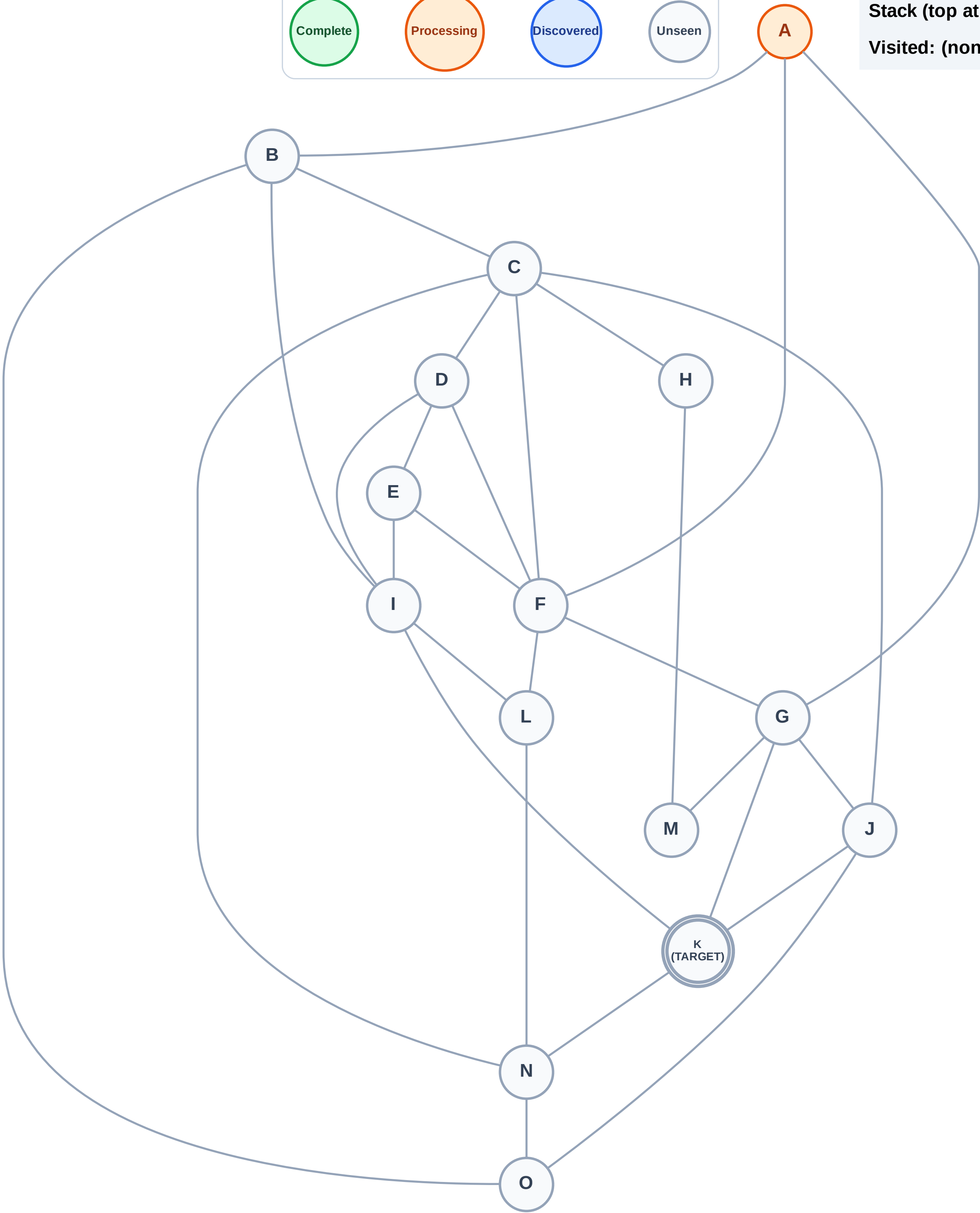
Processing

Discovered

Unseen

Stack (top at right): (empty)

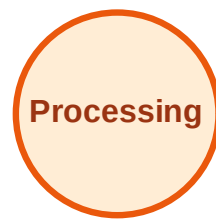
Visited: (none)



DFS Traversal

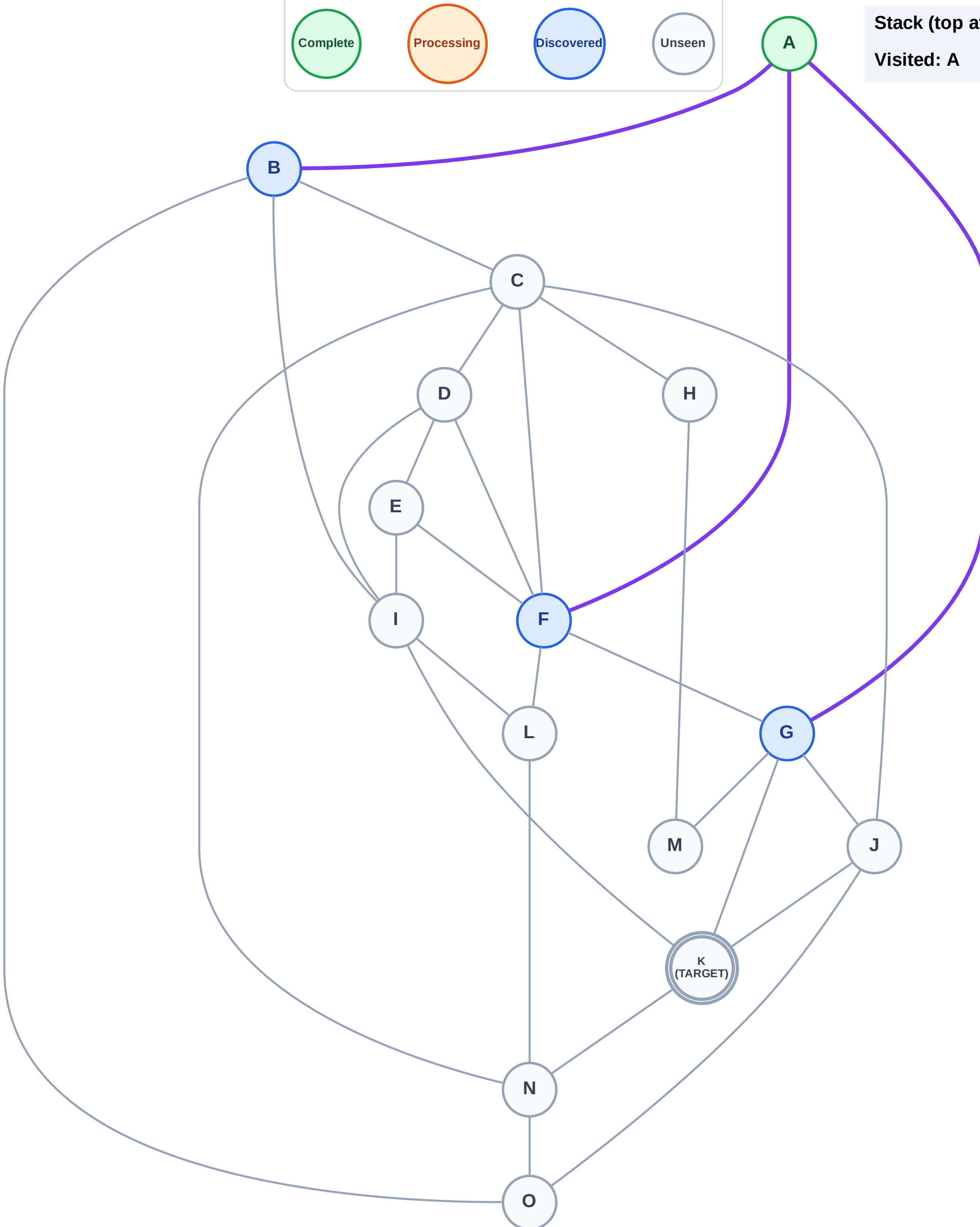
Step 3: Discover G, F, B from A

Legend



Stack (top at right): G | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

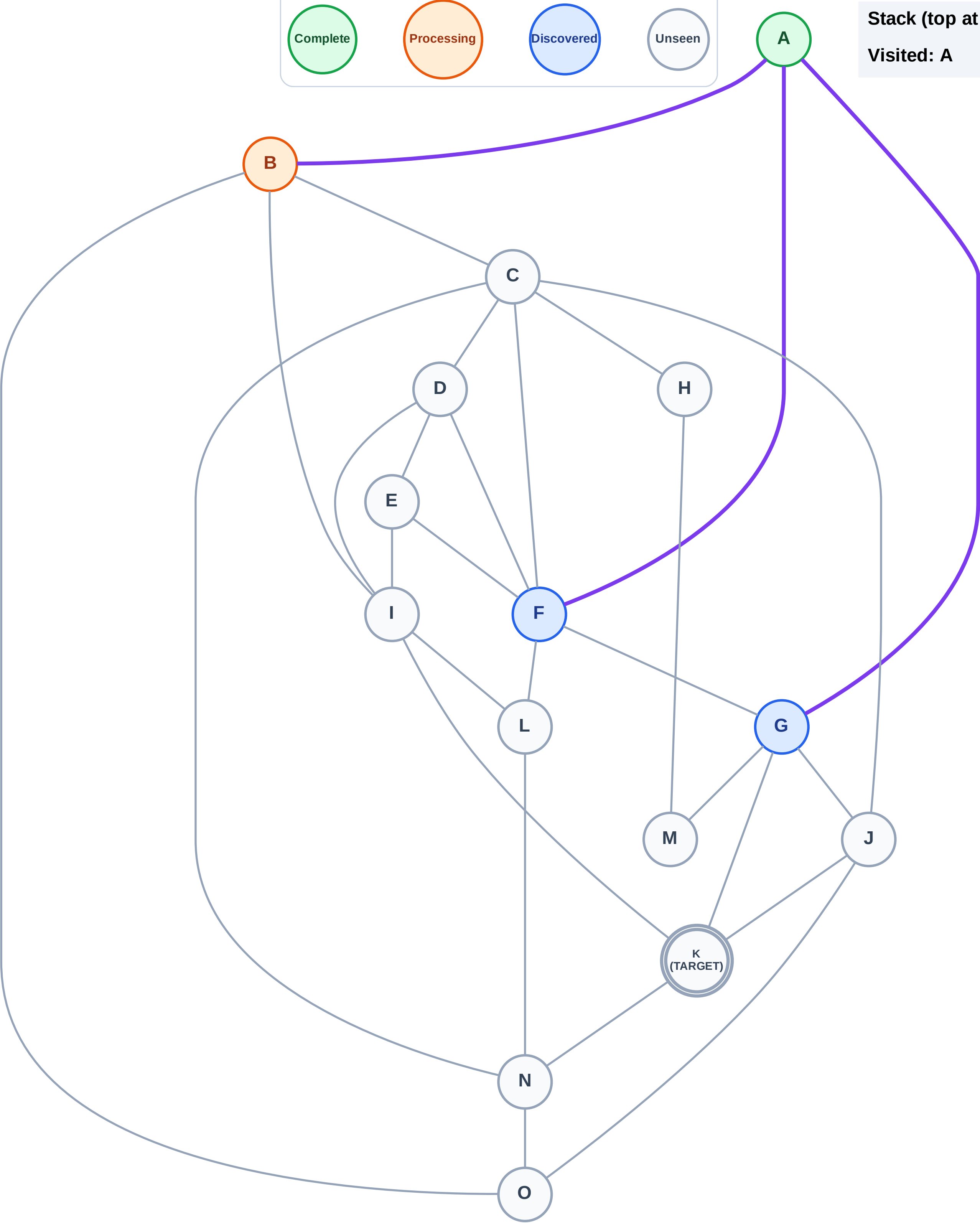
Processing

Discovered

Unseen

Stack (top at right): G | F

Visited: A



Step 5: Discover O, I, C from B

Step 5: Discover O, I, C from B

Legend

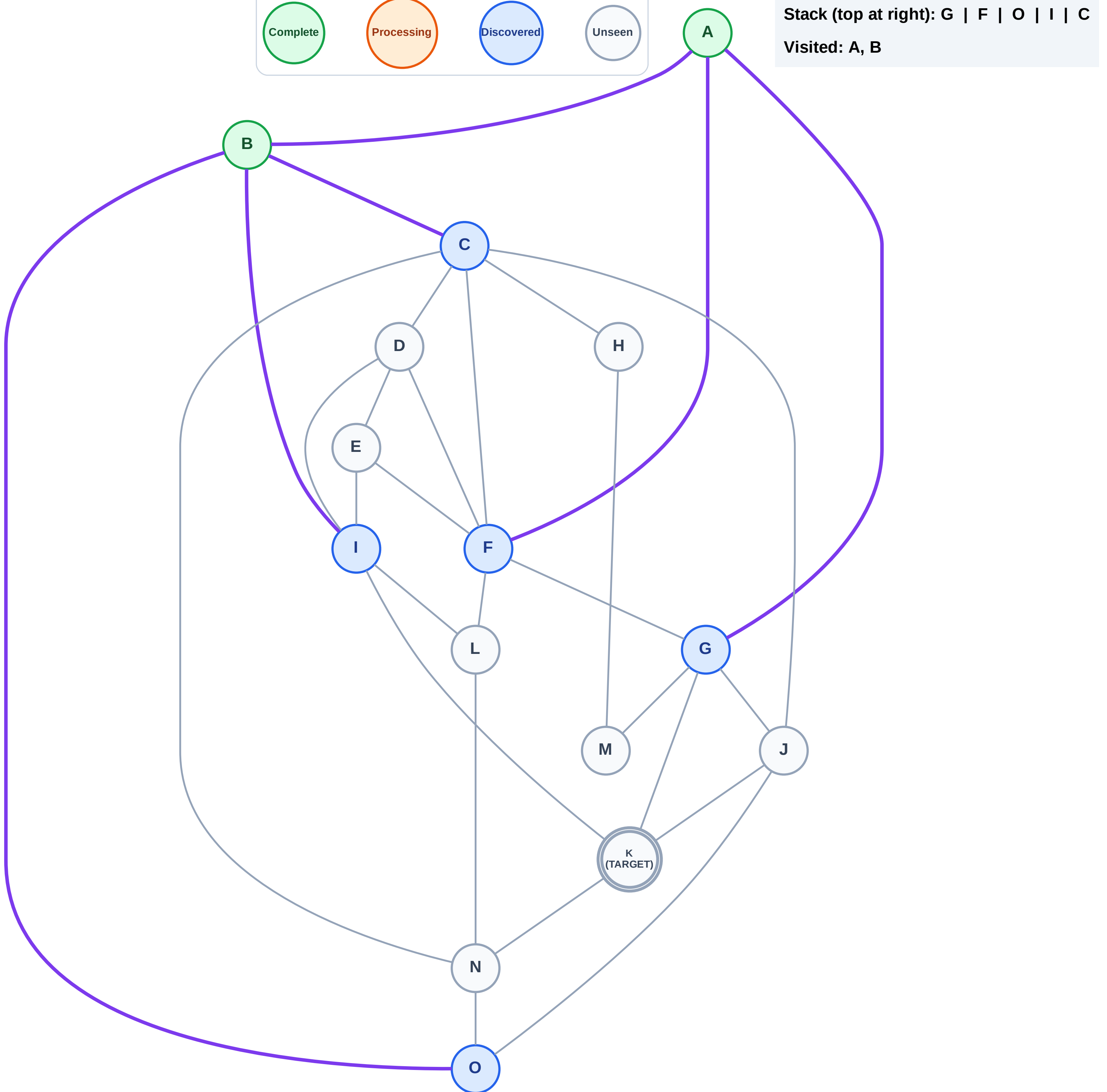


Complete Processing Discovered Unseen

Processing

Unseen

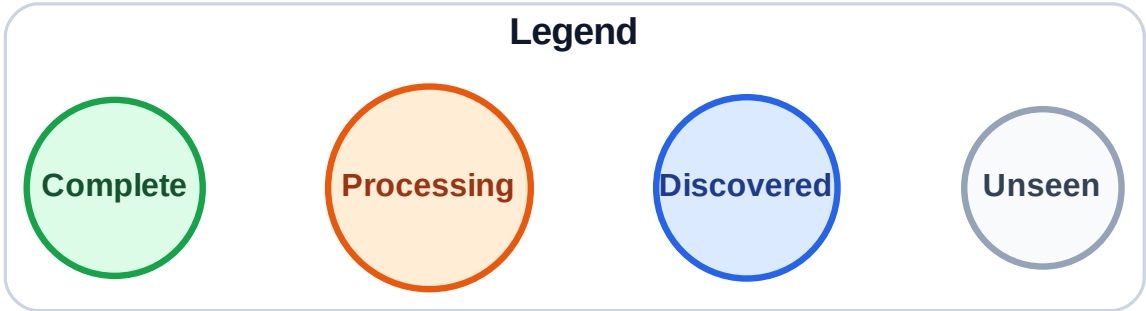
Visited: A, B



DFS Traversal

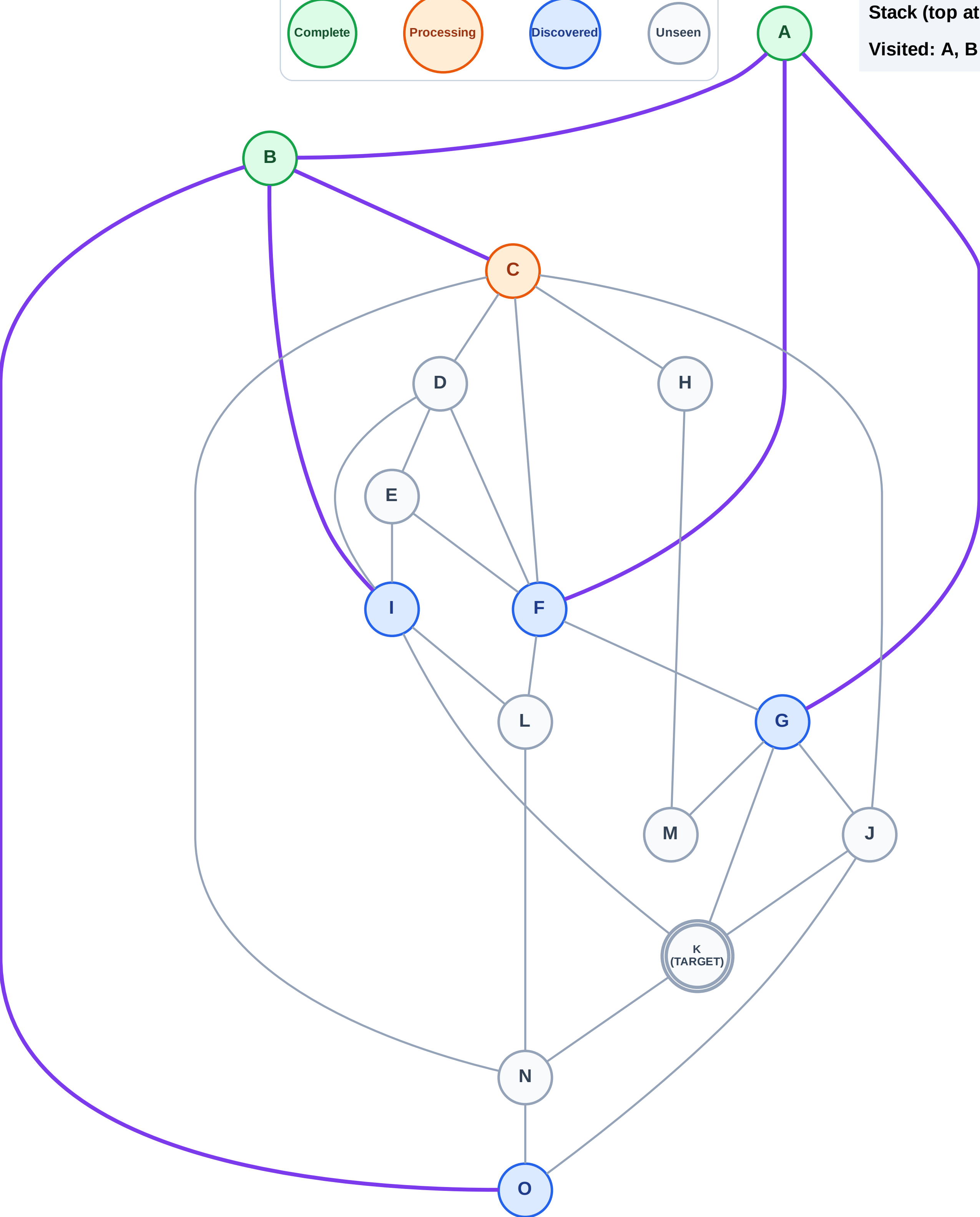
Step 6: Process node C

Legend



Stack (top at right): G | F | O | I

Visited: A, B



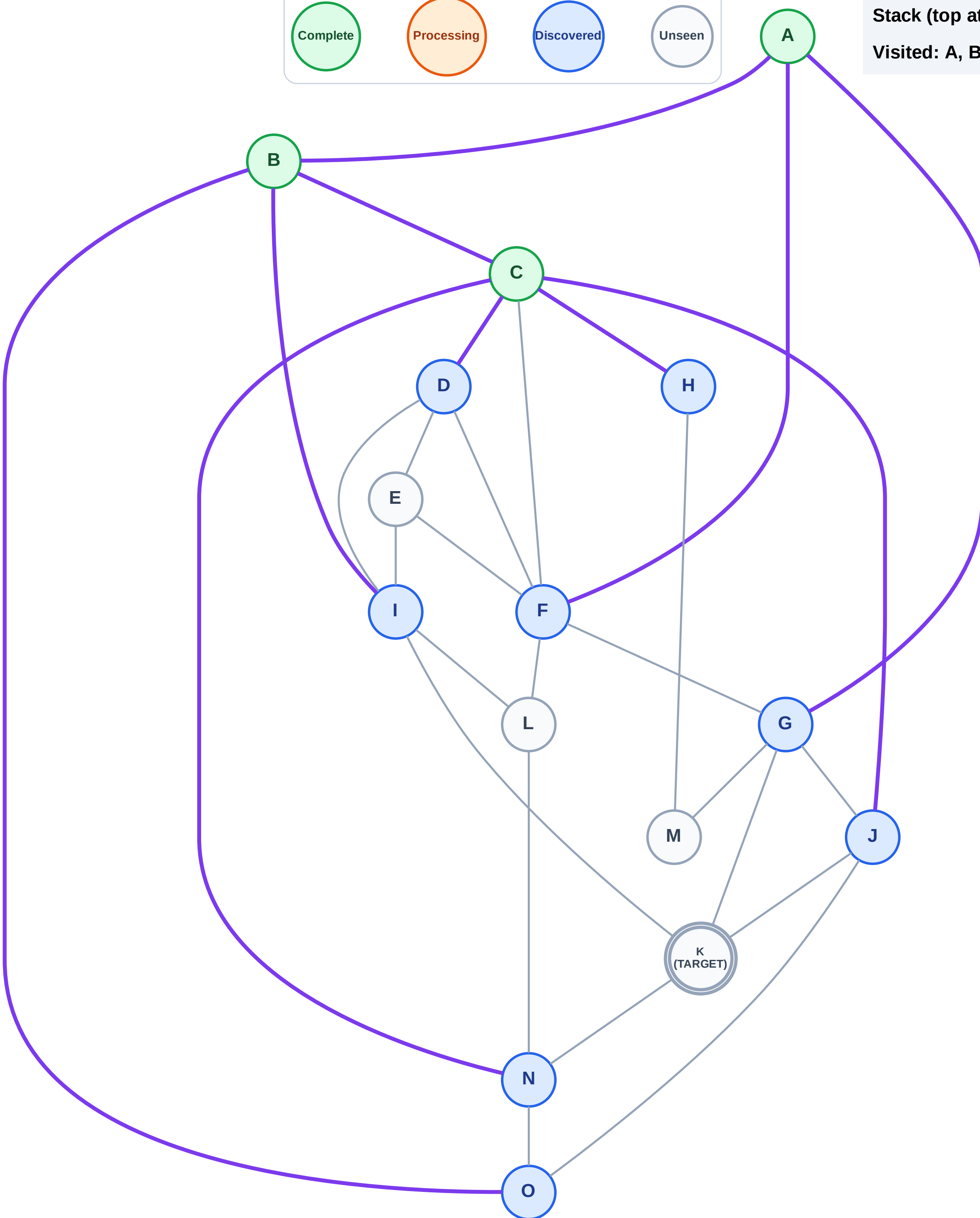
DFS Traversal

Step 7: Discover N, J, H, D from C

Legend



Stack (top at right): G | F | O | I | N | J | H | D
Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

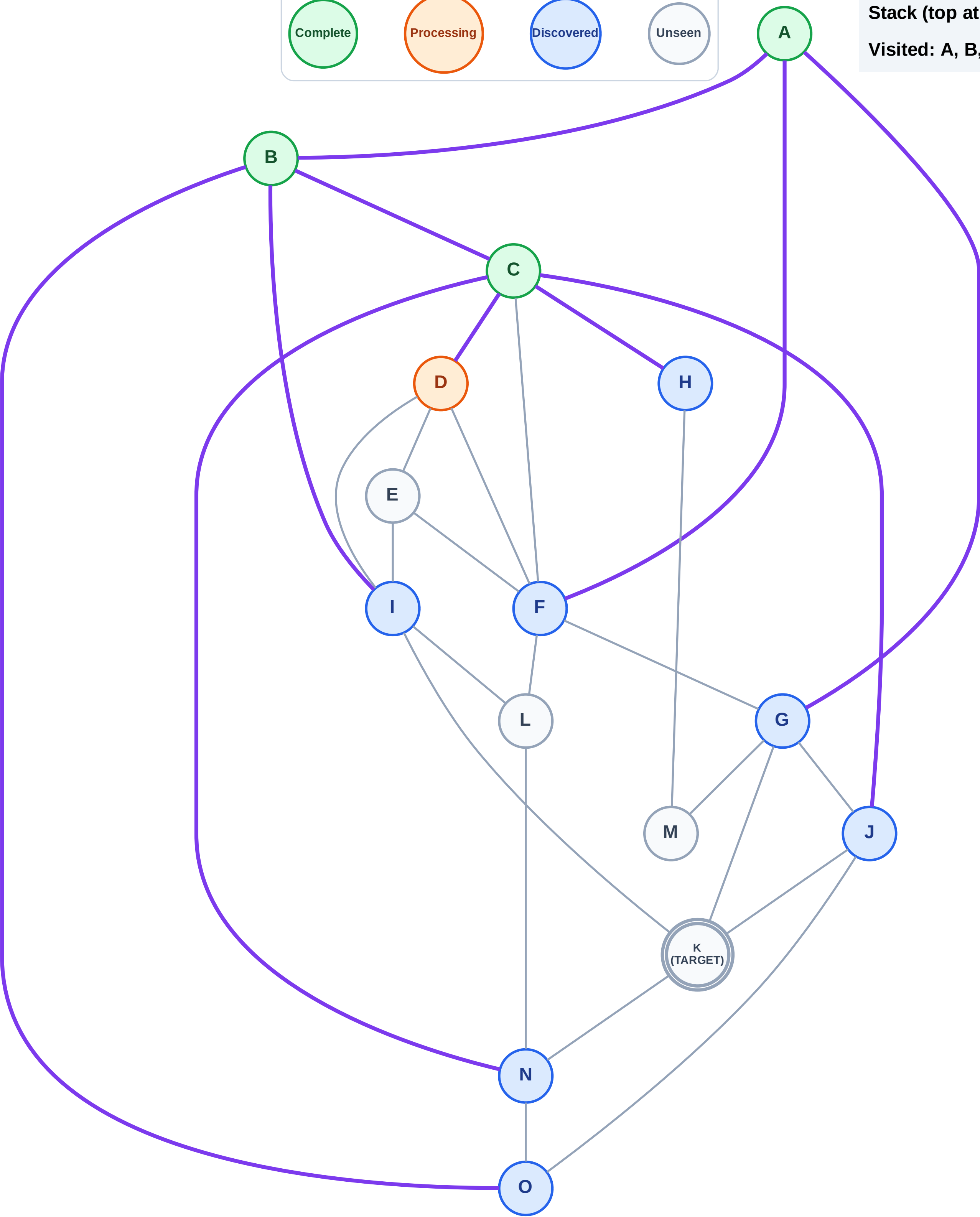
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J | H

Visited: A, B, C



DFS Traversal

Step 9: Discover E from D

Legend

Complete

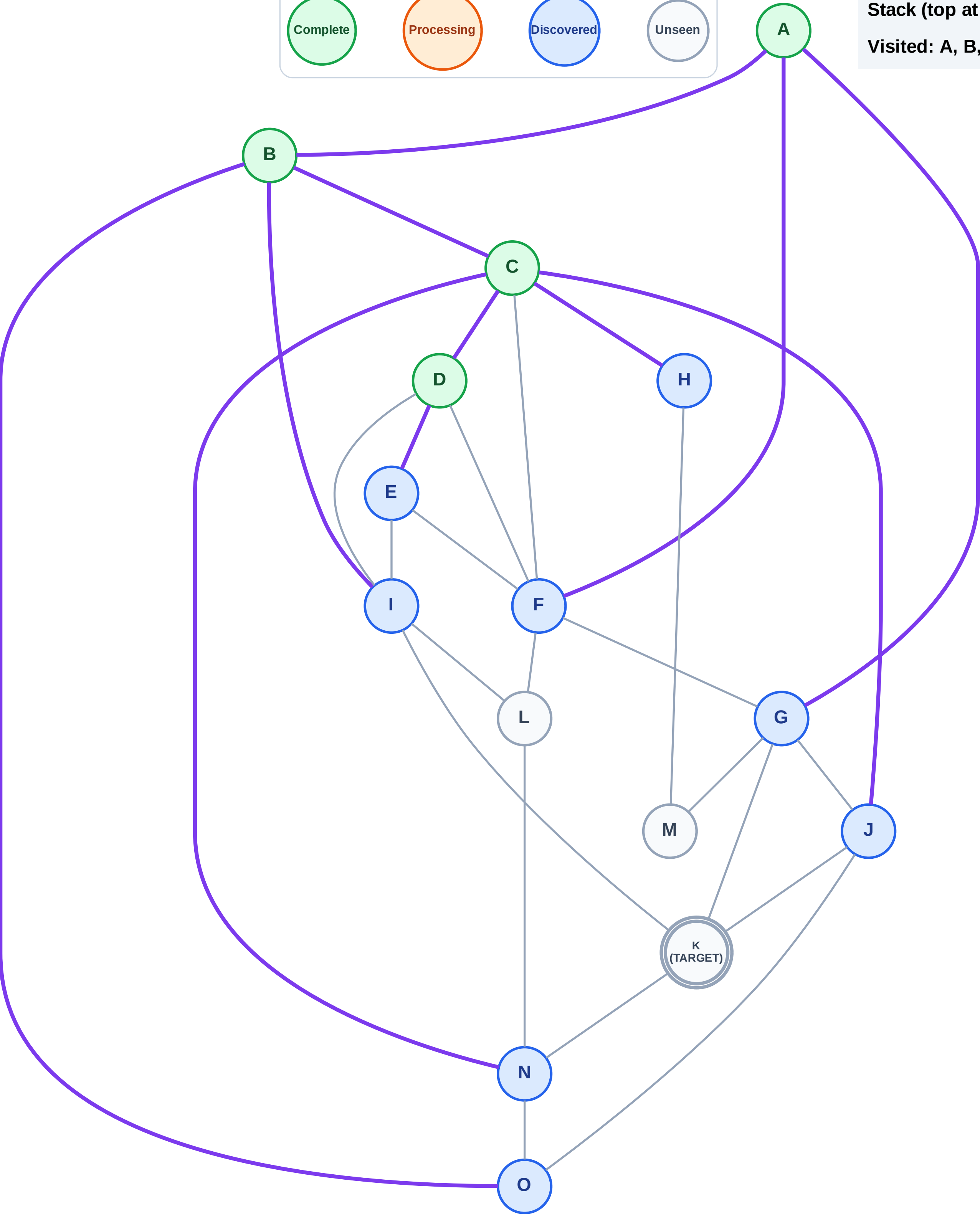
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J | H | E

Visited: A, B, C, D



DFS Traversal

Step 10: Process node E

Legend

Complete

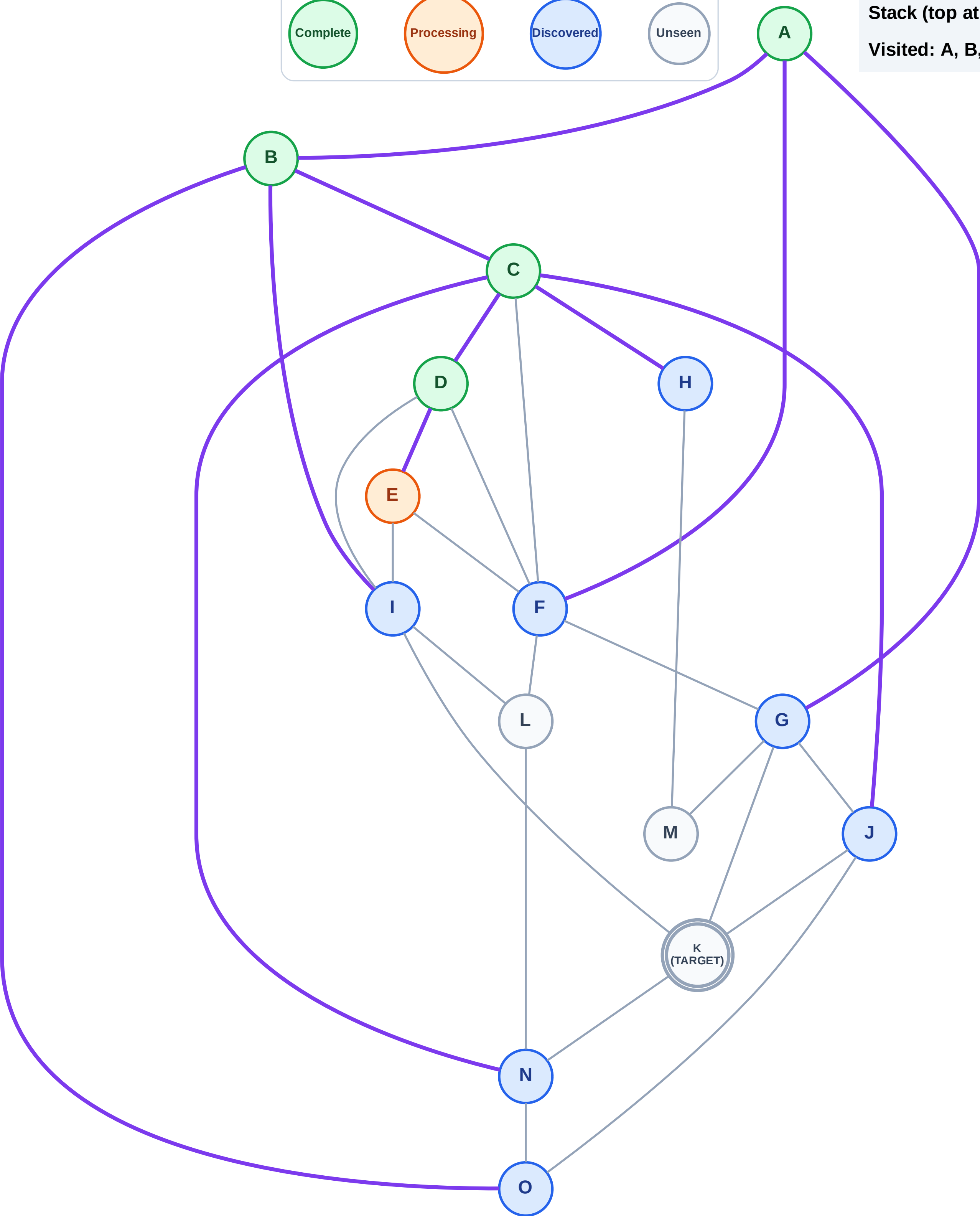
Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N | J | H

Visited: A, B, C, D



DFS Traversal

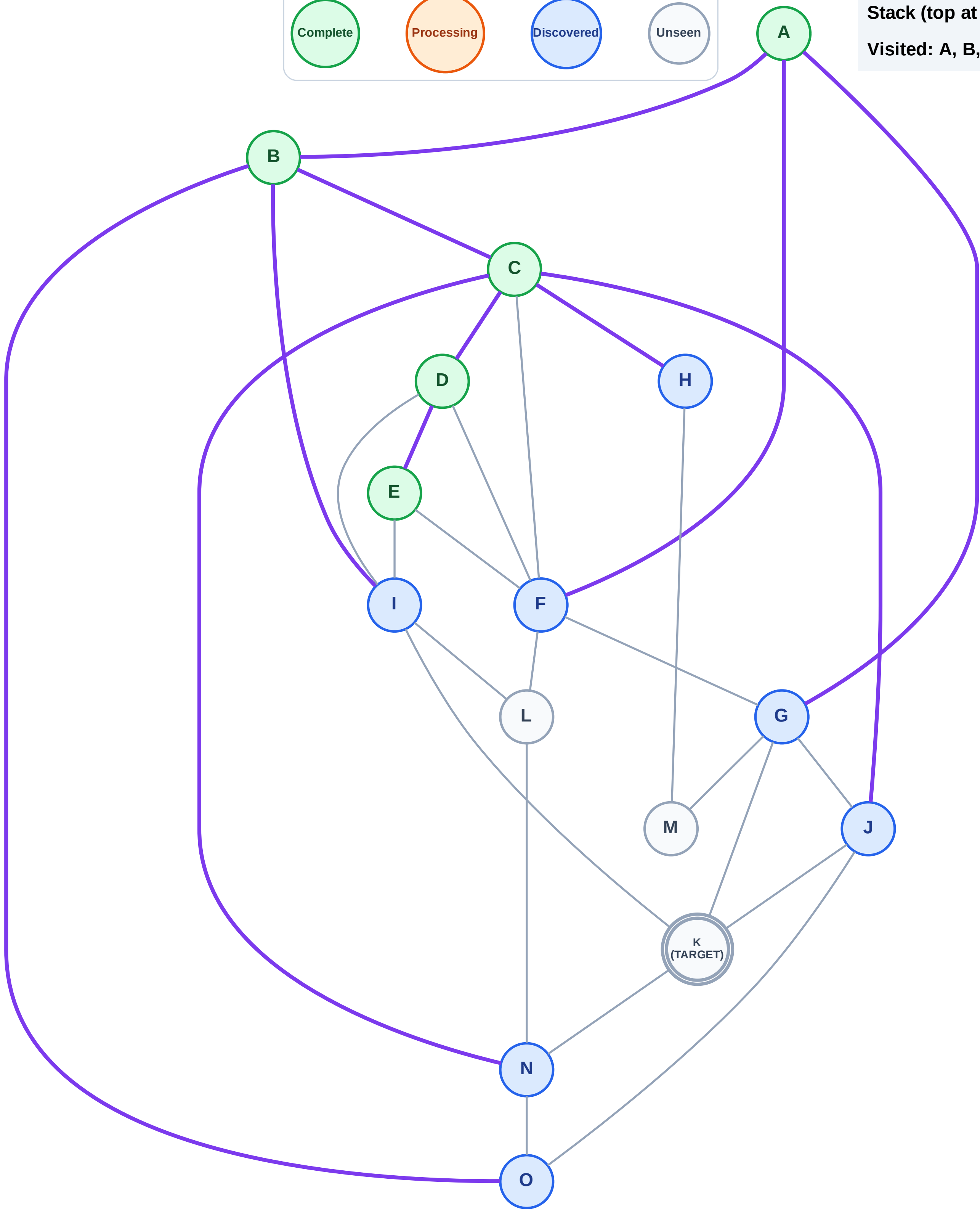
Step 11: No new neighbors from E

Legend



Stack (top at right): G | F | O | I | N | J | H

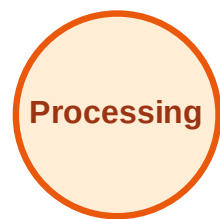
Visited: A, B, C, D, E



DFS Traversal

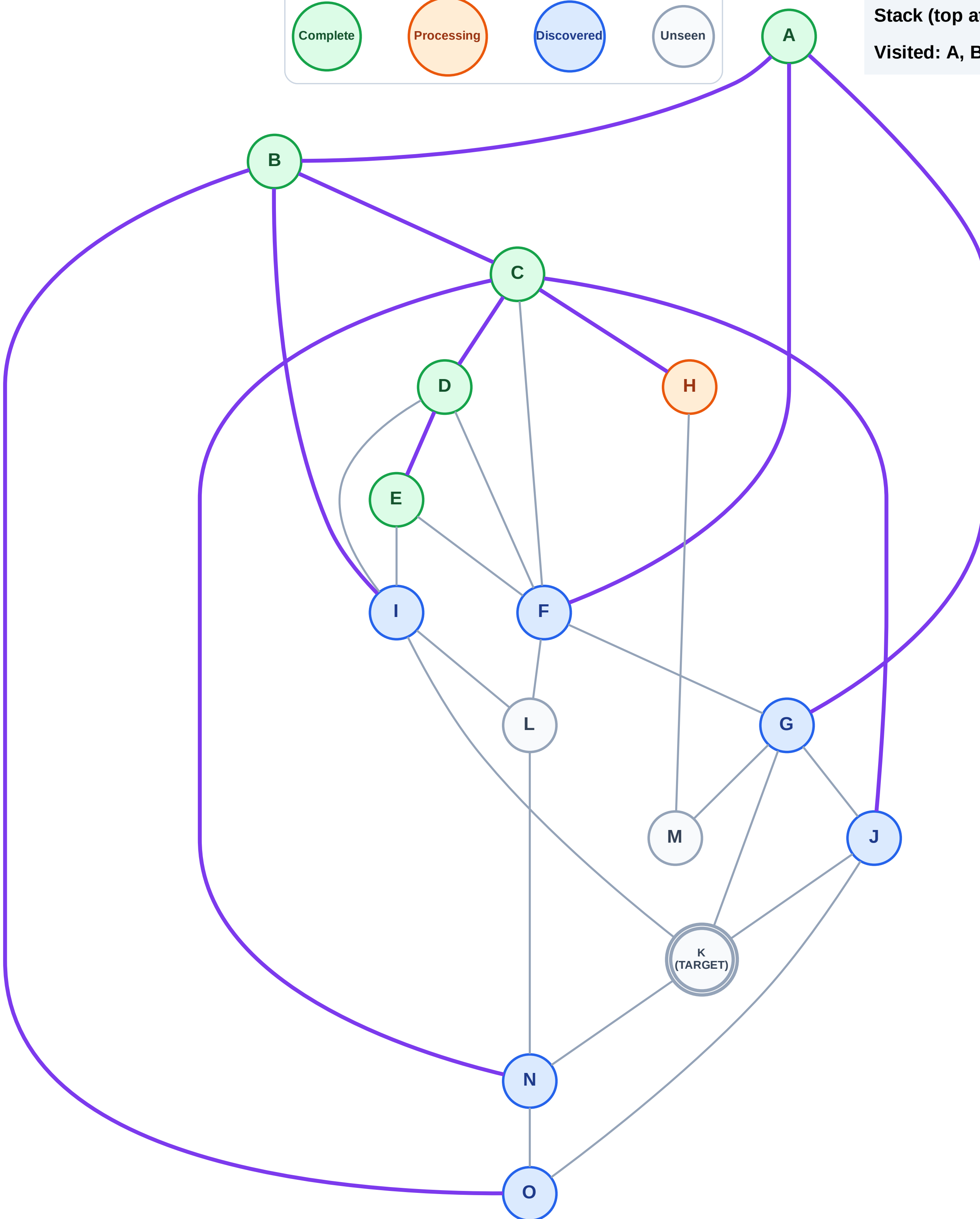
Step 12: Process node H

Legend



Stack (top at right): G | F | O | I | N | J

Visited: A, B, C, D, E



DFS Traversal

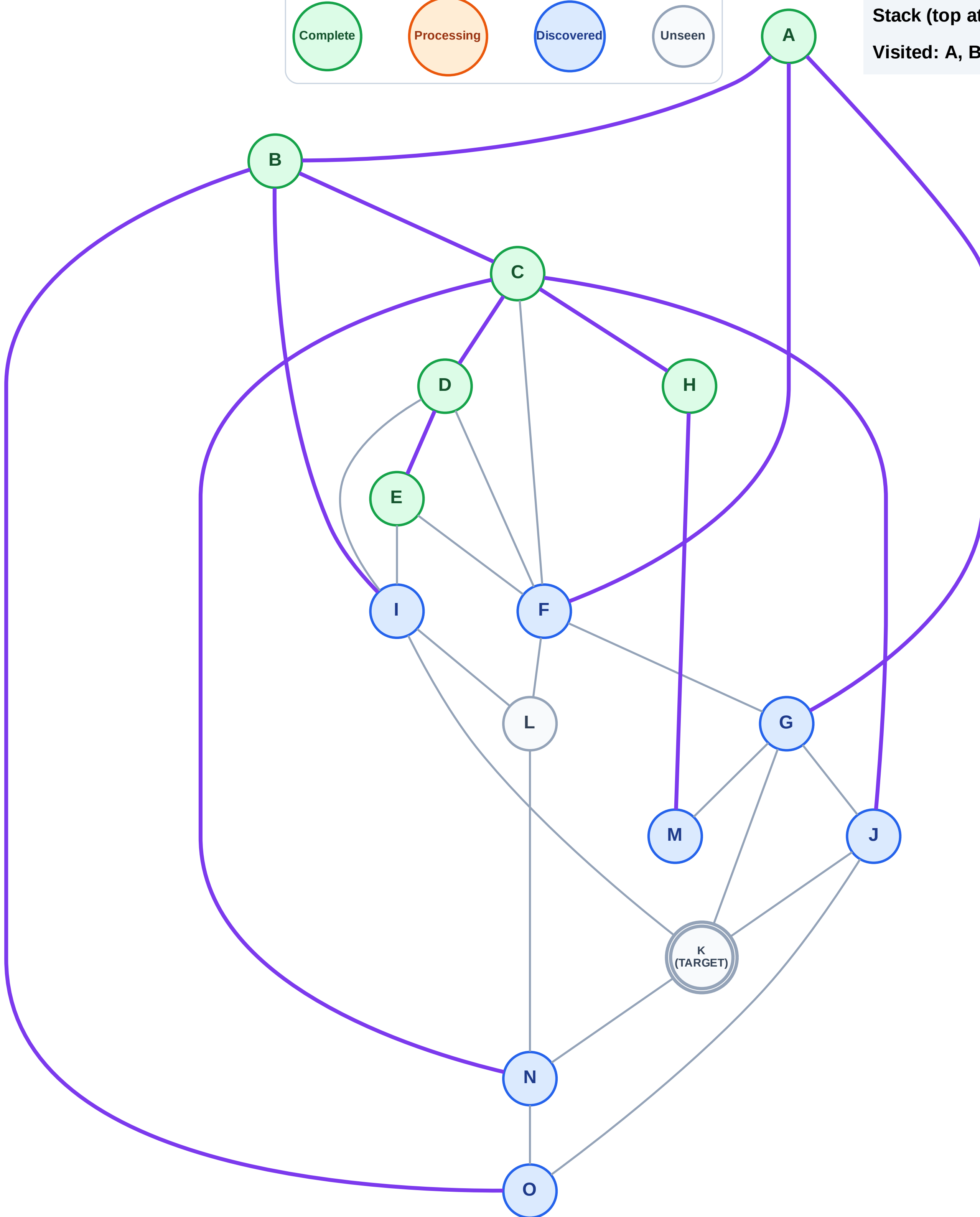
Step 13: Discover M from H

Legend



Stack (top at right): G | F | O | I | N | J | M

Visited: A, B, C, D, E, H



Step 14: Process node M

Legend

Unseen

Visited: A, B, C, D, E, H

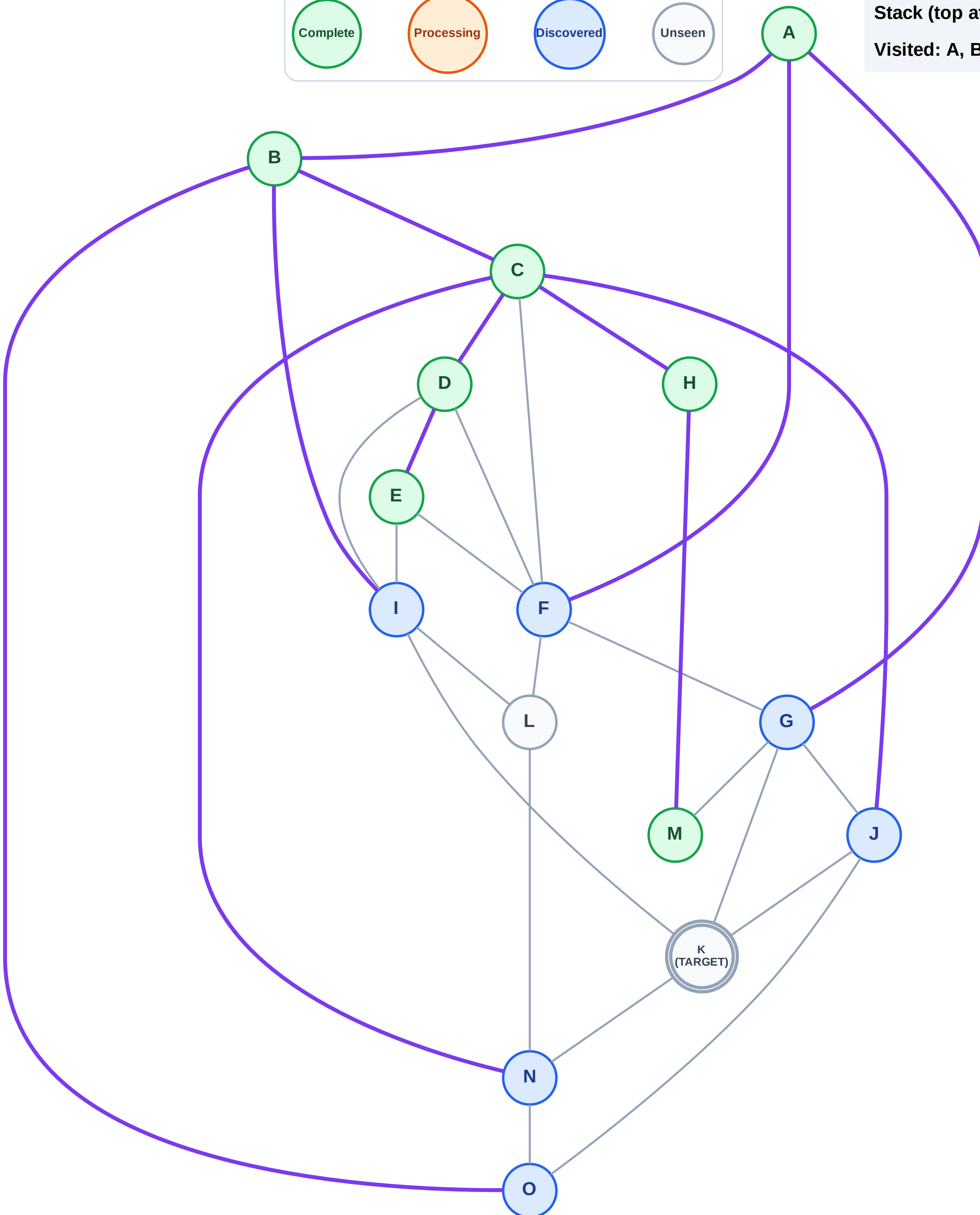
Step 15: No new neighbors from M

Legend

Complete Processing Discovered Unseen

Stack (top at right): G | F | O | I | N | J

Visited: A, B, C, D, E, H, M



DFS Traversal

Step 16: Process node J

Legend

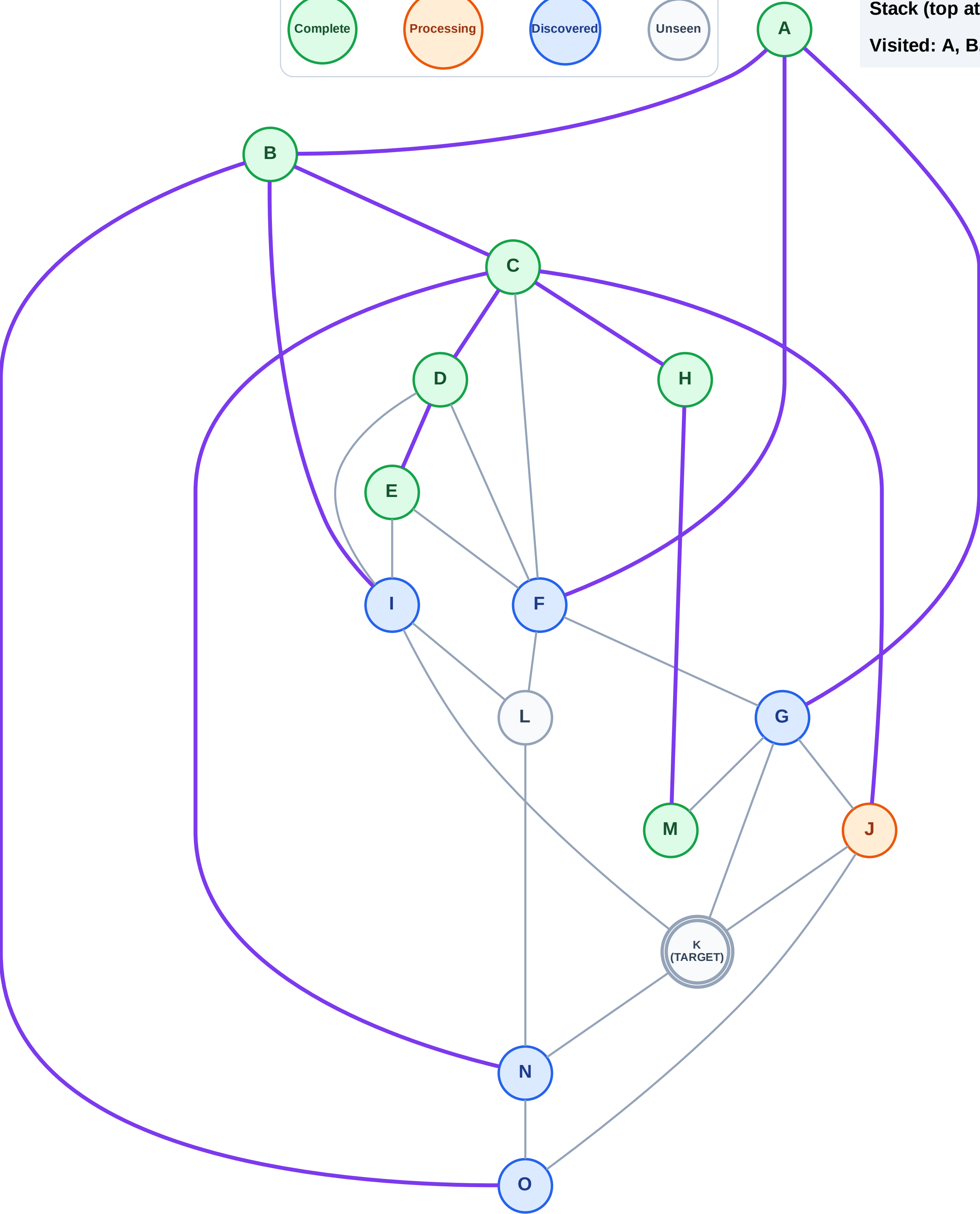
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N
Visited: A, B, C, D, E, H, M



DFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | O | I | N
Visited: A, B, C, D, E, H, J, K, M

